

LIBERO-Object and LIBERO-Spatial: Detailed Results & Failure Analysis

1. Suite-Level Results Summary

Object reaches **99.8% success (499/500)** with a flat, near-ceiling per-task distribution and a single, isolated failure. Spatial reaches **97.4% success (487/500)** with a similarly high average, but the residual error is concentrated in two tasks rather than spread evenly across the suite. Table 1 summarizes both suites side by side; Sections 2 and 3 provide the per-suite detail.

Metric	LIBERO-Object	LIBERO-Spatial
Suite success rate	99.8% (499/500)	97.4% (487/500)
Clean (first-try) success	96.8% (484/500)	95.8% (479/500)
Likely-recovery success	3.0% (15/500)	1.6% (8/500)
Fail timeout	0.2% (1/500)	2.6% (13/500)
Fail other	0% (0/500)	0% (0/500)
Tasks at 100% success	9 / 10	8 / 10
Hardest task	Task 4 (ketchup → basket), 98%	Task 5 (bowl on ramekin), 82%

Table 1. Suite-level outcome summary, 500 episodes per suite.

2. LIBERO-Object: Detailed Findings

2.1 Per-Task Success Breakdown

9 of 10 Object tasks reach 100% success. In **LIBERO-Object Task 4** (ketchup → basket) at 98% (1/50 episodes). No other task shows any failures, so risk in this suite is not distributed across a family of related tasks, it is a single episode within a single task.

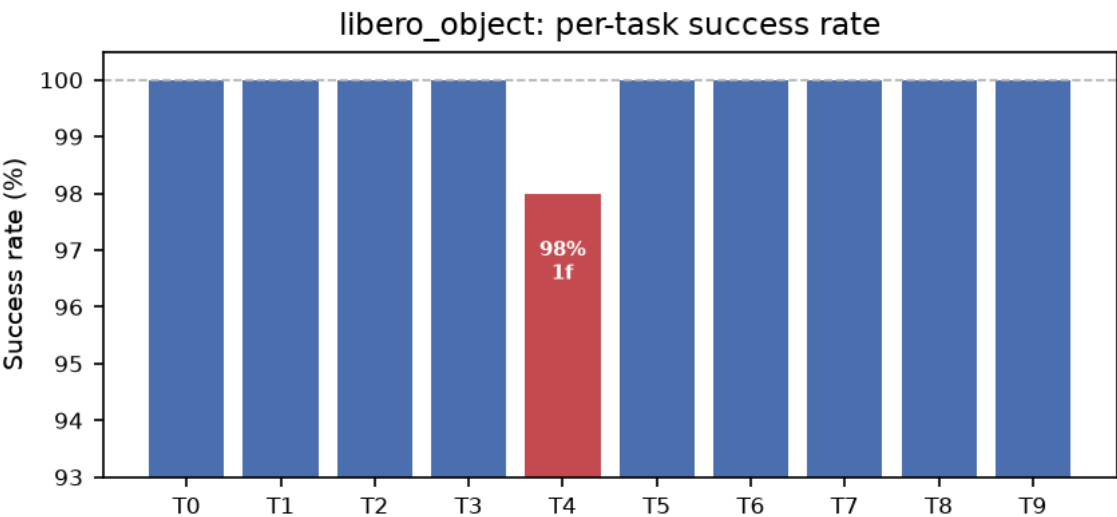


Figure 1. LIBERO-Object per-task success rate. Nine of ten tasks are perfect; Task 4 is the sole outlier at 98%.

2.2 Outcome Composition

Of 500 episodes, 484 (96.8%) are clean, first-try successes, 15 (3.0%) are “likely-recovery” successes (elevated grasp count and/or longer trajectories that still succeed) and 1 (0.2%) is a timeout failure. There are zero “other” failures. Episode length for successes clusters around ~130 steps, while the one failure sits at the max_steps ceiling of 280.

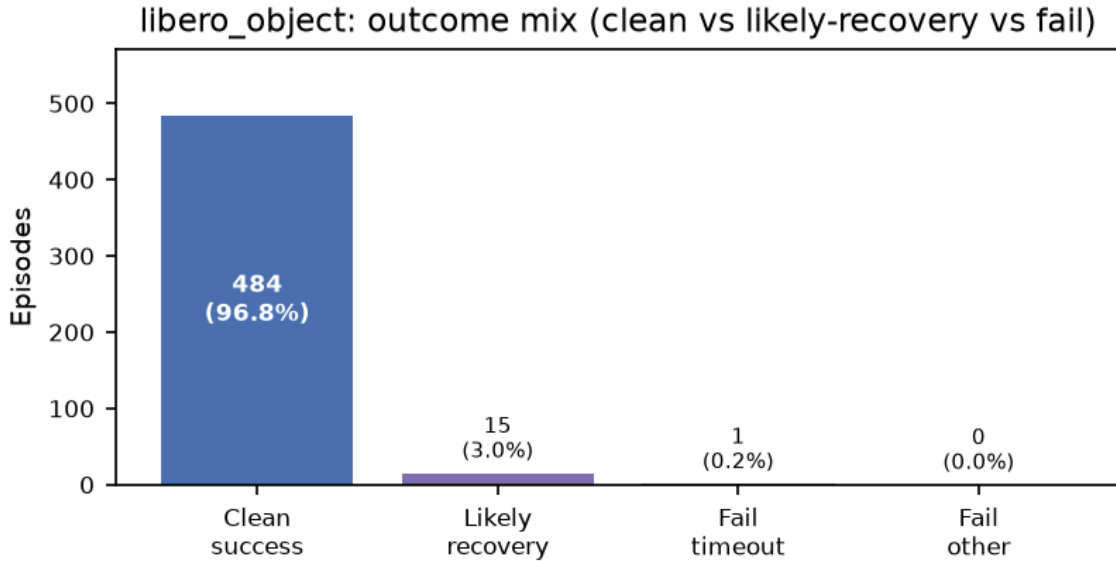


Figure 2. LIBERO-Object outcome mix 484 clean successes, 15 likely-recovery successes, 1 timeout failure, 0 other failures.

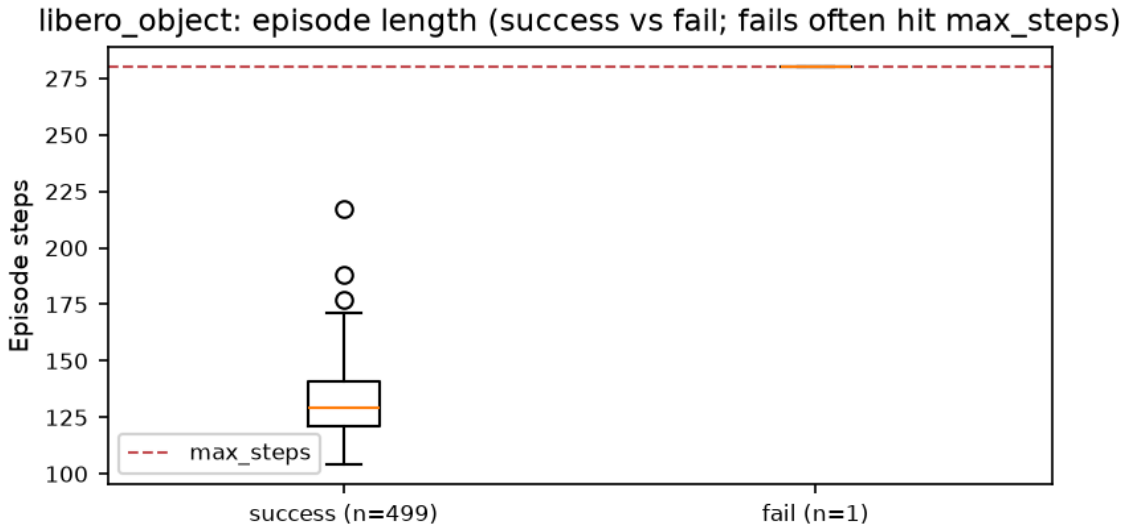


Figure 3. Episode length, success vs. failure. Successes complete near ~130 steps; the single failure runs to the 280-step limit.

2.3 The Single Failure Case: Mechanism and Evidence

The LIBERO-Object Task 4 failure (ketchup → basket) is characterized by the following per-episode evidence:

- **Correct object at first close:** the first gripper closure is on the correct target.
- **Elevated grasp count:** 5 grasp attempts, versus 0 to 1 for the large majority of successful episodes.
- **Non-trivial pick displacement (≈ 0.20):** the object is measurably lifted, so the grasp is at least partially engaged.
- **Large final place distance (≈ 0.31 m):** the object ends far from the basket, versus 0.01–0.09 m for successes.
- **Outcome:** matches the heuristic tagger's single “place_issue” label (0 wrong_object / long_path / other tags), consistent with a stalled placement rather than a wrong grasp.

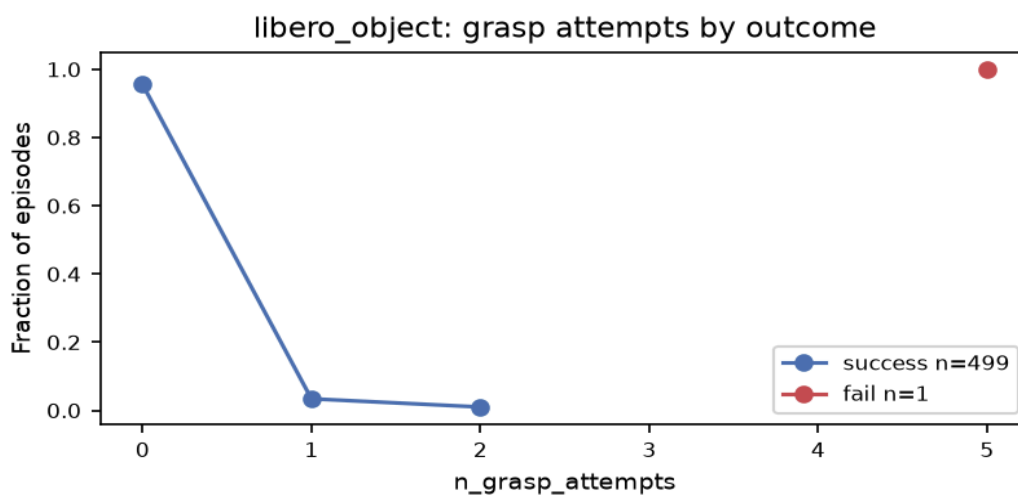


Figure 4. Grasp attempts by outcome. Successes cluster at 0 to 1 grasps; the failure required 5.

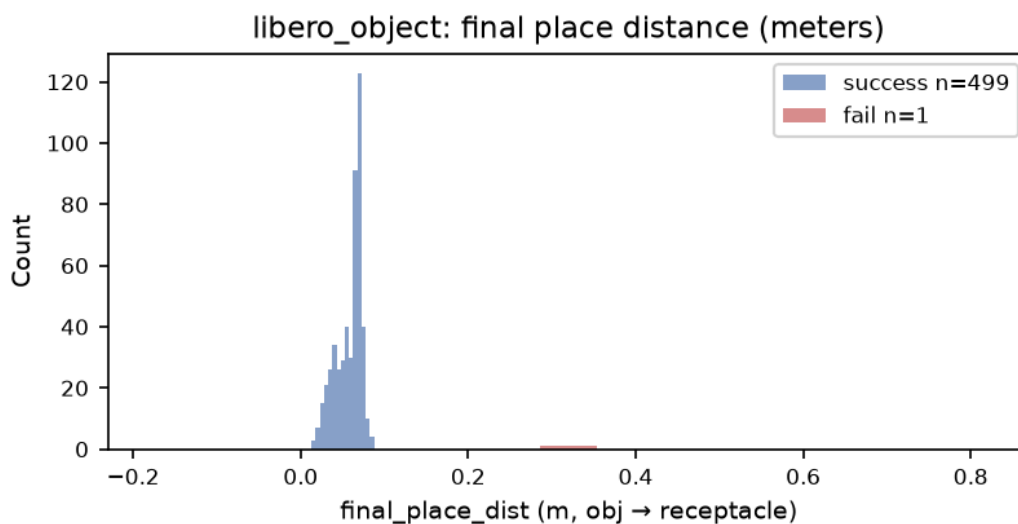


Figure 5. Final place distance. Successes converge within centimeters of the receptacle; the failure ends ~0.32 m away.

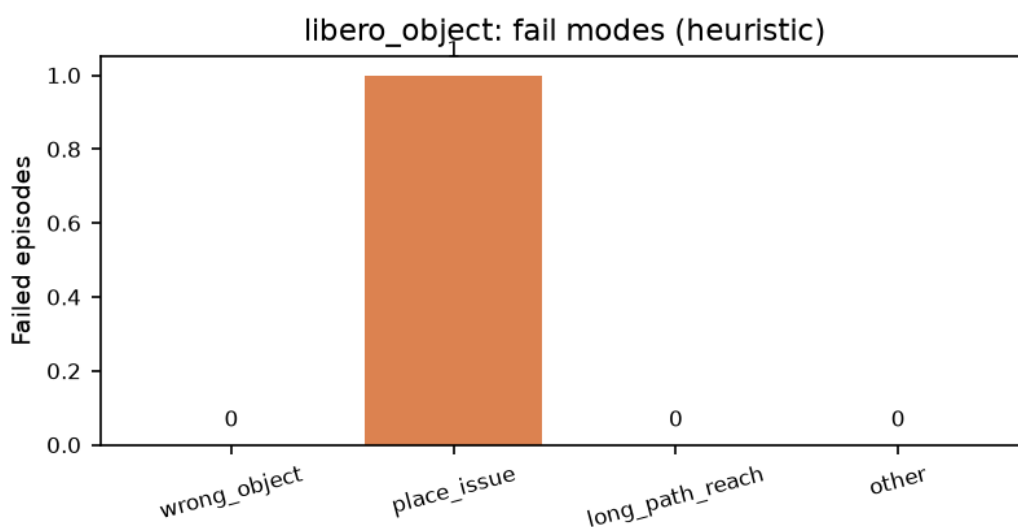


Figure 6. Heuristic fail-mode tags. The single failure is tagged `place_issue`; `wrong_object`, `long_path`, and `other` are all zero.

2.4 What This Rules In and Rules Out

Wrong-object-at-first-close is approximately 0% across all 500 Object episodes, and the one failure has a **correct** first close. This rules out a visual-identification or object-grounding explanation for the residual error. The convergent evidence correct grasp target, partial lift, repeated re-grasping, and a large uncorrected place distance instead localizes the failure to **post pick place completion**: the policy does not reliably finish delivering an object to its receptacle once the initial pick attempt is imperfect. Identity-conditioned manipulation (selecting the correct item among distractors) is not implicated as a bottleneck in this suite.

On Object, the suite is near-saturated (99.8%), and that residual error looks like a place/completion issue, so the more relevant pressure point is the **action/control pathway** rather than the vision model for identity. Behavior is still largely first-try: about **96.8% clean success**, only **~3% likely recovery**, and the single fail hits the step limit after many grasps with the object still far from the receptacle retries without a successful finish.

3. LIBERO-Spatial: Detailed Findings

3.1 Per-Task Success Breakdown

8 of 10 Spatial tasks reach 100% success. The remaining two carry all of the suite's error mass: **LIBERO-Spatial Task 4** (bowl → top drawer → plate) at 92% (4/50 failures) and **Task 5** (bowl → ramekin → plate) at 82% (9/50 failures). Unlike Object, risk here is concentrated in a small, identifiable subset of tasks rather than a single isolated episode.

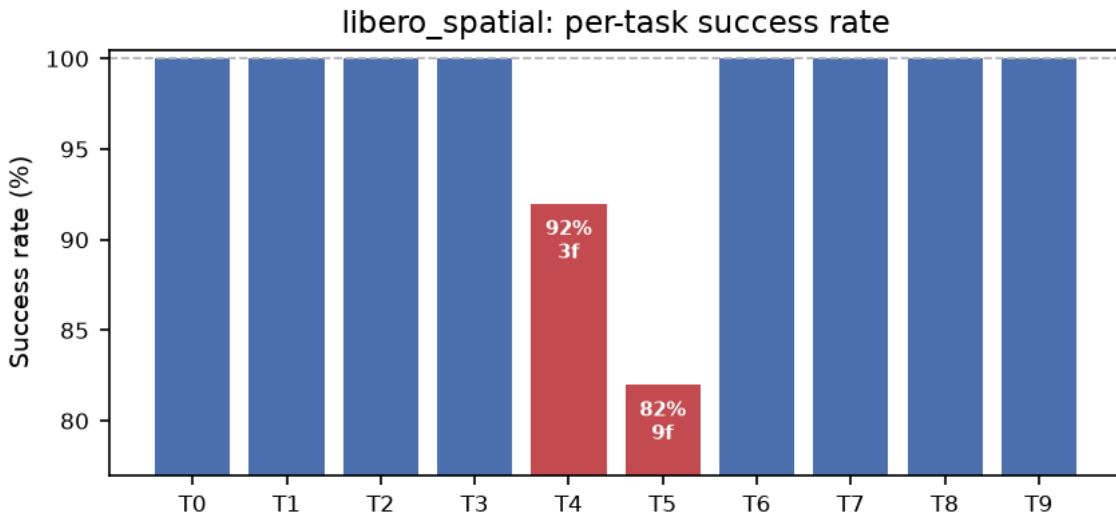


Figure 7. LIBERO-Spatial per-task success rate. Risk concentrates in Task 4 (92%) and Task 5 (82%); the remaining eight tasks are perfect.

3.2 Outcome Composition

Of 500 episodes, 479 (95.8%) are clean successes, 8 (1.6%) are likely-recovery successes and 13 (2.6%) are timeout failures, with zero “other” failures. Episode length for successes clusters around ~100 steps. Notably, **all 13 failures sit at the max_steps ceiling of 280** every failure is a genuine timeout, not an early, decisive miss.

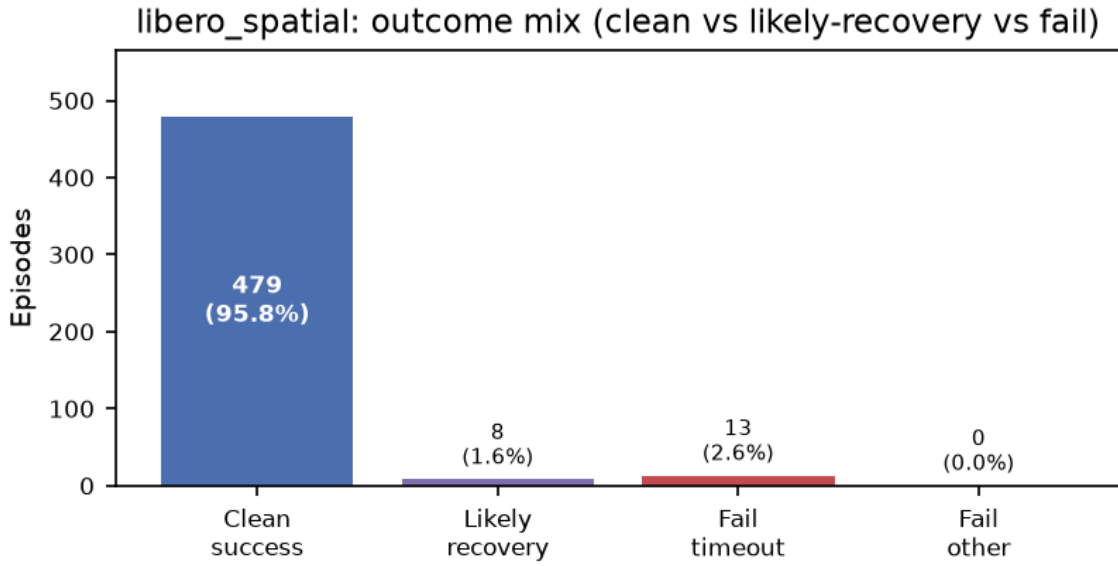


Figure 8. LIBERO-Spatial outcome mix 479 clean successes, 8 likely-recovery successes, 13 timeout failures.

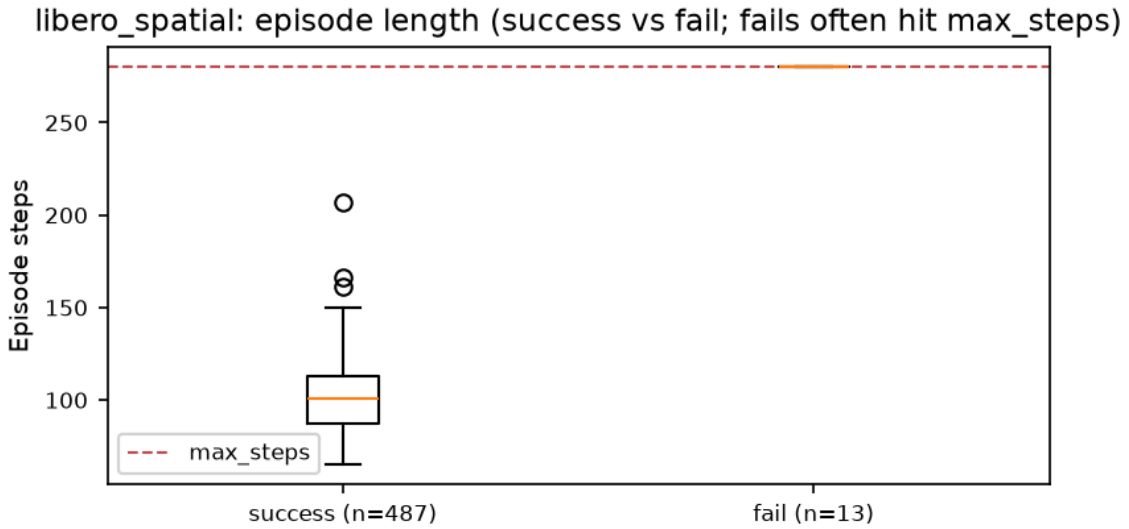


Figure 9. Episode length, success vs. failure. Successes complete near ~100 steps; all 13 failures run to the 280-step limit.

3.3 Failure Cluster A Task 5 (Bowl on Ramekin): Grounding Failure

Task 5 is the weakest task in either suite (82%, 9/50 failures). The wrong-object-at-first-close rate rises sharply on this task, and **7 of the suite's 13 failures (54%)** involve the gripper's first closure landing on the *ramekin* the support object named in the relational phrase rather than the *bowl*, the actual grasp target. Once this mis-grounding occurs, **recovery is 0%**: no failed episode in this cluster subsequently re-targets the correct object, the episode ends with the bowl never grasped.

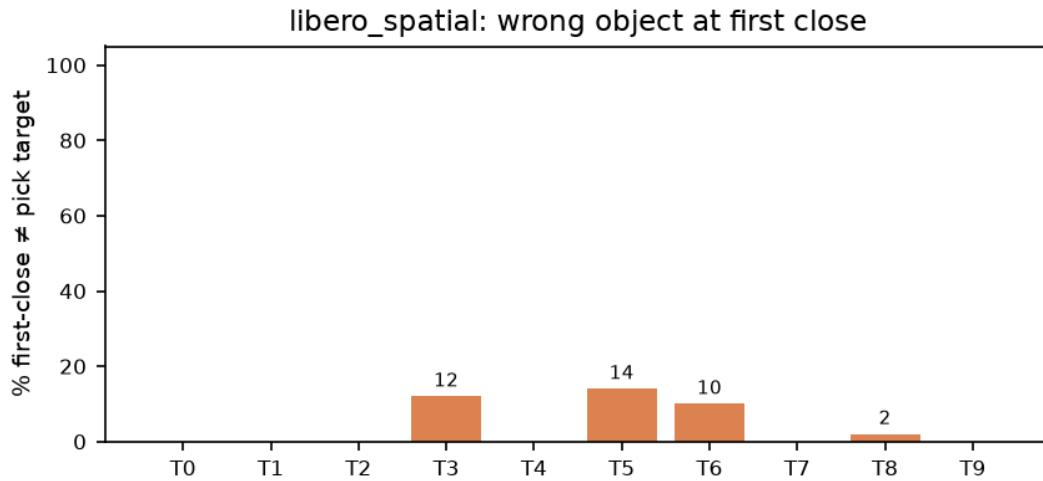


Figure 10. Wrong-object-at-first-close rate by task. Task 5 shows the highest rate (~14%), with secondary elevation on Tasks 3 and 6; Task 4 is ~0%.

3.4 Failure Cluster B Task 4 (Drawer Extraction): Grasp/Control Failure

Task 4 fails at 92% (4/50 episodes), but its wrong-object-at-first-close rate is approximately 0% the policy correctly identifies and closes on the bowl. Its failures instead split into:

- **Grasp/control (4/13 suite failures, 31%):** correct first close, but low pick displacement relative to successful episodes and repeated grasp attempts without a stable lift.
- **Reach/targeting (2/13 suite failures, 15%):** the end-effector never converges close enough to the target to initiate a clean grasp.

Both sub-modes are control-side rather than identity-side: the policy “knows” which object to grasp but struggles to execute the reach, insertion, and lift required by the drawer’s constrained geometry.

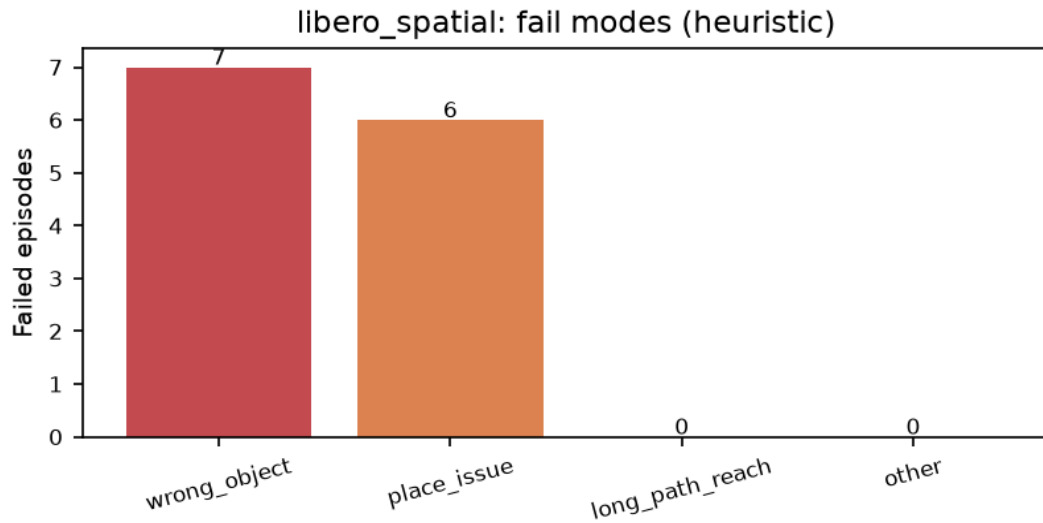


Figure 11. Fail-mode taxonomy for the 13 LIBERO-Spatial failures: wrong-object/grounding (7/13, 54%) concentrates in Task 5; grasp/control and reach/targeting (6/13, 46%) concentrate in Task 4.

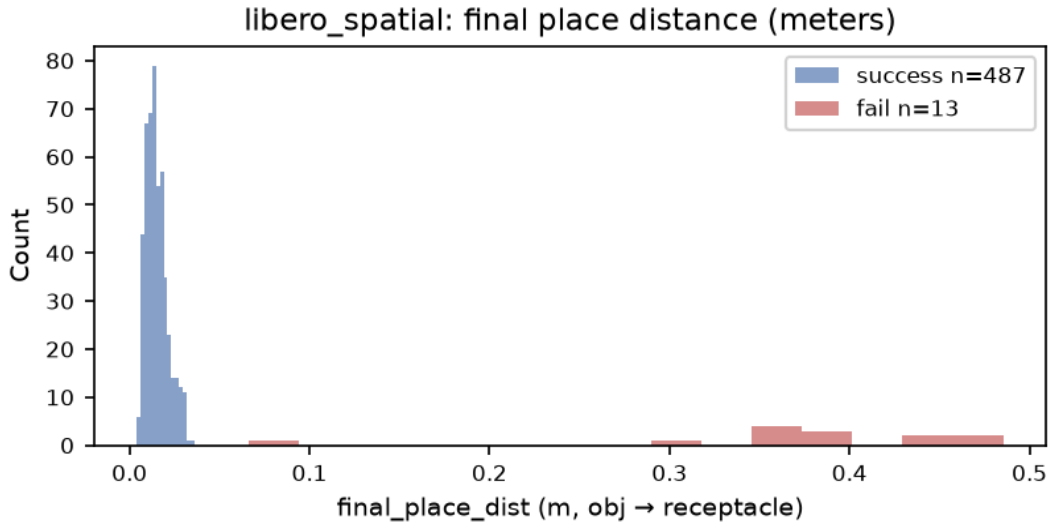


Figure 12. Final place distance, LIBERO-Spatial. Successes end within centimeters of the receptacle ($\sim 0.01\text{--}0.04\text{ m}$); failures typically end far away ($\sim 0.3\text{--}0.5\text{ m}$), consistent with neither cluster reaching a completed placement.

3.5 What This Rules In and Rules Out

Most spatial-relation phrasings in the suite (next to, between, from the center, in the drawer) succeed at or near 100%, which rules out a general failure to parse spatial language. The two hard tasks instead separate cleanly along different axes: **LIBERO-Spatial Task 5** is a grounding/vision limitation, while **Task 4** is a control limitation despite correct object identification. Both share one property: once a failure state is reached, the policy does not correct course, it continues attempting the task rather than recovering.

On Spatial, residual failures are concentrated in a few tasks and often involve wrong object at first close, which points to the **vision–language grounding pathway** how MolmoAct2 binds camera views and the instruction to the correct object among spatial distractors. We still describe behavior as largely first-try here: about **95.8%** of episodes are clean successes, only **~1.6%** look like likely recovery, successful runs usually use one grasp, and all 13 fails time out at max steps even when the policy makes extra grasp attempts, rather than clearly correcting and finishing.

4. Cross-Suite Comparison

4.1 Where the Suites Agree

- Both suites are dominated by clean, first-try successes (96.8% Object, 95.8% Spatial).
- In both suites, every recorded failure is a step-limit timeout, no early aborts, no other failure types.
- In both suites, “likely-recovery” successes are mostly extra grasps on an already-correct object, not genuine correction of an earlier error.

4.2 Where the Suites Diverge

Dimension	LIBERO-Object	LIBERO-Spatial
Distribution of risk	Single isolated episode (Object T4)	Concentrated in two tasks (Spatial T4, T5)
Wrong-object-at-first-close	$\sim 0\%$ suite-wide	Elevated on Spatial T5 ($\sim 14\%$)
Dominant failure locus	Post pick place/completion (control)	Grounding (Spatial T5) and grasp/control (Spatial T4)

Dimension	LIBERO-Object	LIBERO-Spatial
Recovery after mis-grounding	N/A no mis-grounding observed	0% (Spatial T5)

5. Takeaways

LIBERO-Object is effectively solved for instance-level visual targeting: the policy reliably selects the correct object among distractors, and its one residual failure is a control-side place-completion issue rather than a perception error.

LIBERO-Spatial is strong on average but not uniform: its error mass concentrates in two individually diagnosable tasks **Spatial Task 5** dominated by a grounding/vision limitation, **Spatial Task 4** by a control limitation rather than a general weakness in following spatial instructions.

Jointly, the two suites indicate that this checkpoint behaves as a high-success, first-try policy with limited closed-loop correction: it performs strongly when the first attempt matches its training distribution, and it does not reliably recover once an early error a wrong grasp target or a failed lift pushes an episode out of that distribution.